Natural Language Processing : Gender Classification and Political affiliation detection

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1 Introduction

The goal of this project is to analyze a corpus of manifestos of candidates in the elections of the French fifth republic using Natural Language Processing (NLP) techniques. We want to check if there is some semantic similarity between documents regarding candidates of the same gender or political party, try and identify latent political structures in the corpus, then visualize documents in a low-dimensional semantic space which will help us detect outlier documents that are too far from mainstream political speeches. All the work can be found in `https://github.com/KingDonatien/NLP_donatien_roi`.

2 Collecting and visualizing data

2.1 Dataset

The dataset that we will use consists of approximately 20,000 scanned manifestos of candidates in the elections of the French fifth republic (from around 1970–1980) collected by Sciences Po. Each document includes OCR-extracted text and metadata such as year, candidate name, political party, and political support.

2.2 Preprocessing

We decided to apply the following preprocessing steps:

- OCR text cleaning and normalization
- Removal of punctuation and non-informative symbols
- Standardization of text encoding

so that finding semantic membership of a text becomes easier in what we will do next.

2.3 Descriptive Statistics

The corpus has the following characteristics:

- Strong class imbalance in political parties (a lot of texts belong to PS)
- High variability in document length due to OCR noise
- Presence of structured headers (e.g., election labels, institutions)

We respectively visualize in **Figure 1** and **Figure 2** the distribution of the corpus in terms of text length and gender. It was to be expected that around 80% of the corpus is masculine, and we also see that the majority of the texts seems to belong to PS (Parti Socialiste).

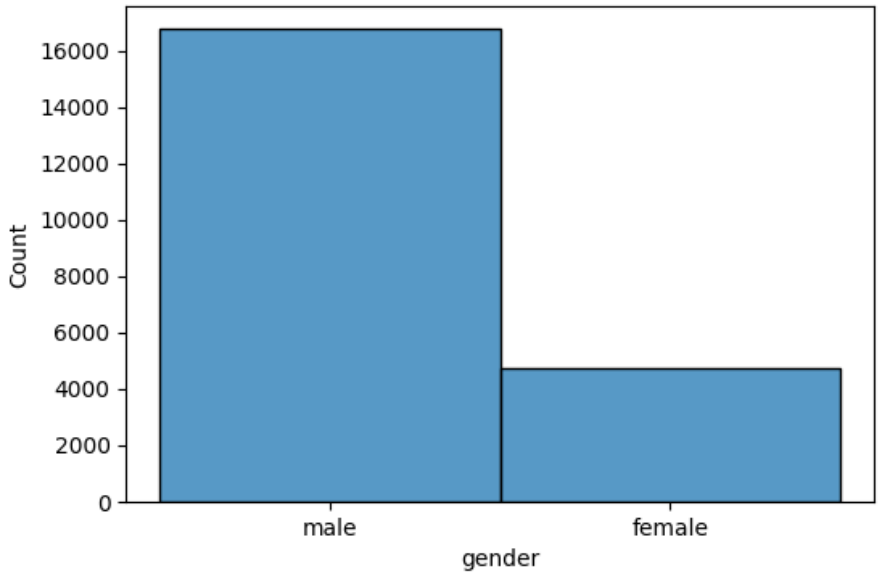

Figure 1: Distribution of Text length

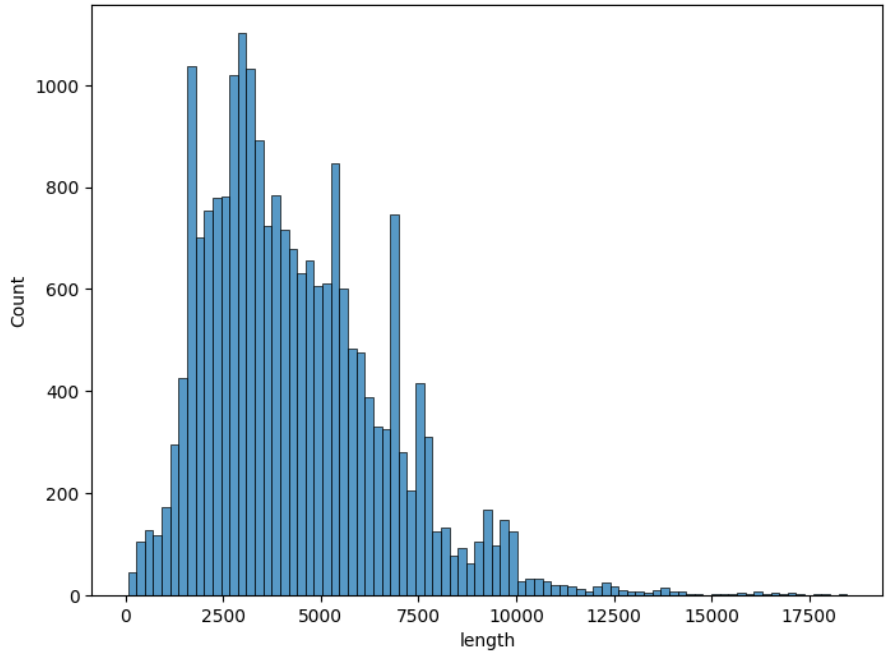

Figure 2: Distribution of Gender

3 A quick review of the state-of-the-art

Semantic similarity in NLP is commonly modeled using vector representations of text, where semantic relationships are encoded in geometric distances between embeddings (1).

Recent advances in transformer-based models such as Sentence-BERT allow documents to be represented as dense vectors capturing contextual meaning (2). These embeddings can then be analyzed using dimensionality reduction techniques such as UMAP or t-SNE (3).

A standard pipeline for embedding-based document analysis includes:

1. Embedding generation
2. Dimensionality reduction
3. Clustering
4. Visualization and interpretation

Outlier detection has also been shown to be useful for identifying anomalous or noisy documents in embedding spaces.

4 Our Methodology

In this section, we will go through each step of our methodology, explaining what we are going to do and why, while staying quite concise.

4.1 Document Embeddings

Each document d_i is encoded into a dense vector representation: $x_i = f(d_i), \quad x_i \in R^n$

where f is a pretrained multilingual sentence transformer model. It is a quite simple model to understand and it has clearly sufficient potential for what we want to do next.

4.2 Dimensionality Reduction

To enable visualization, embeddings are projected into a two-dimensional space using UMAP:

$$y_i = \text{UMAP}(x_i), \quad y_i \in R^2$$

UMAP preserves both local and global structure of the semantic space, so that we don't lose much of the structure just for the sake of visualization.

4.3 Clustering and Metadata Analysis

Now, all documents are analyzed according to features we try to predict, using a MultinomialNB classifier in all cases:

- Gender of the candidate
- Political party affiliation

Thus, we can expect that clustering in embedding space reveals latent ideological structures that depend on the political affiliation and/or the gender of the candidate.

4.4 Outlier Detection

Outliers are identified using a really simple distance-based scoring in the embedding space, for example the Euclidean distance. We can expect the outliers to correspond to OCR errors, atypical political content, or hybrid documents.

5 Experiments and results

Now that we have explained our methodology, we can proceed to our actual experiments and results. For the embedding, we use a multilingual Sentence Transformer model trained on large-scale parallel corpora to compute semantic embeddings.

5.1 Accuracy of the predictors

Unfortunately, we cannot say that our MultinomialNB predictors we developed for gender and party detection are truly efficient, achieving accuracies of around 0.8, but they are still the best ones,

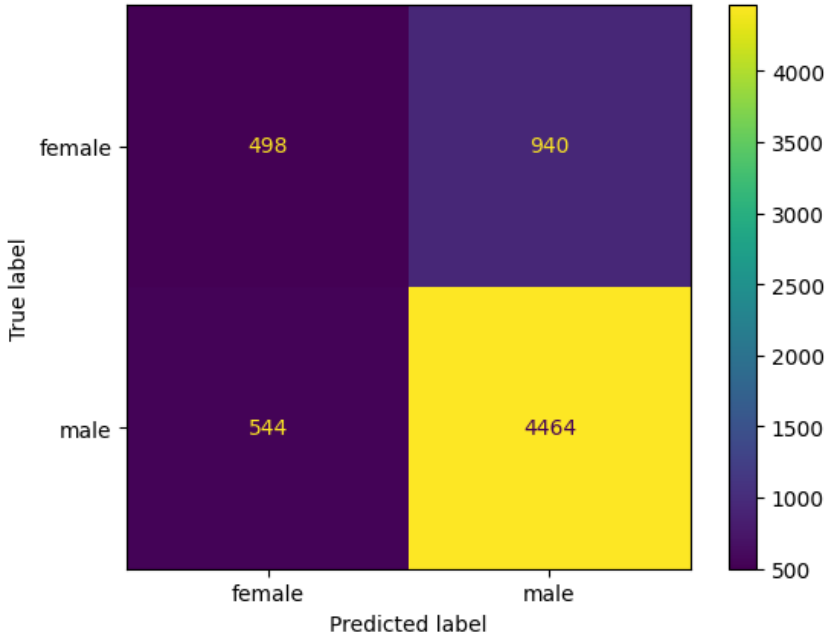

Figure 3: Confusion Matrix of gender predictor (MultinomialNB)

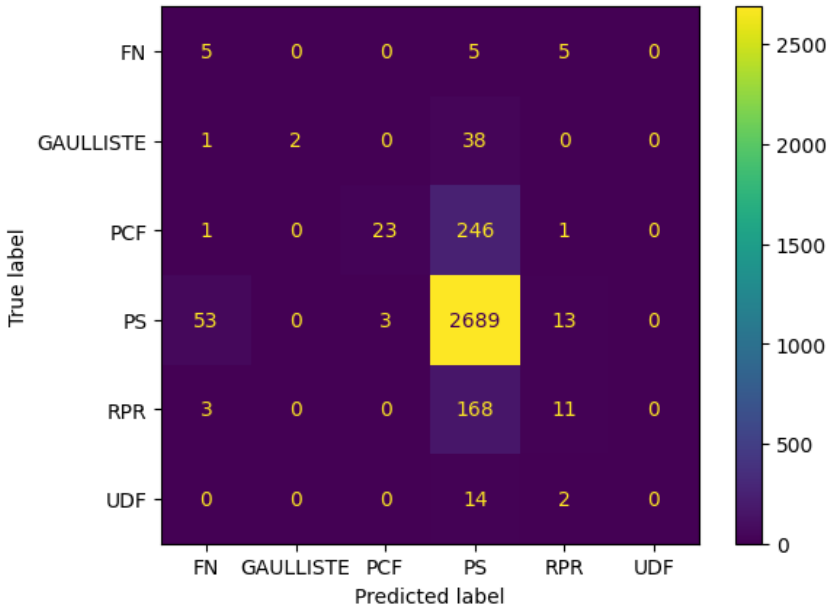

Figure 4: Confusion Matrix of political party predictor (MultinomialNB)

compared to simple Logistic Regressions for example. We respectively display in **Figure 3** and **Figure 4** the confusion matrices related to gender classification and political party classification.

The matrices sadly reveal that a lot of texts that do not belong to the majority class (texts that come from a woman or texts that do not belong to the Parti Socialiste) are misclassified as texts of the majority class.

5.2 Visualization in the embedding space

As we already explained before, we project embeddings into 2D using UMAP and thus we visualize:

- The structure of genders in **Figure 5**
- The structure of political parties in **Figure 6**
- Outliers as the lone points of the previous graphs

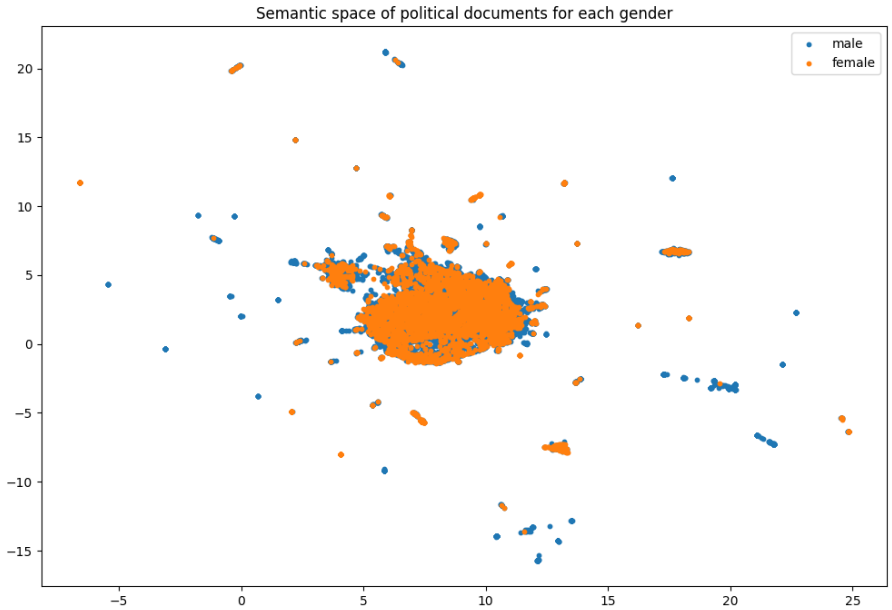

Figure 5: 2D semantic embedding colored by gender

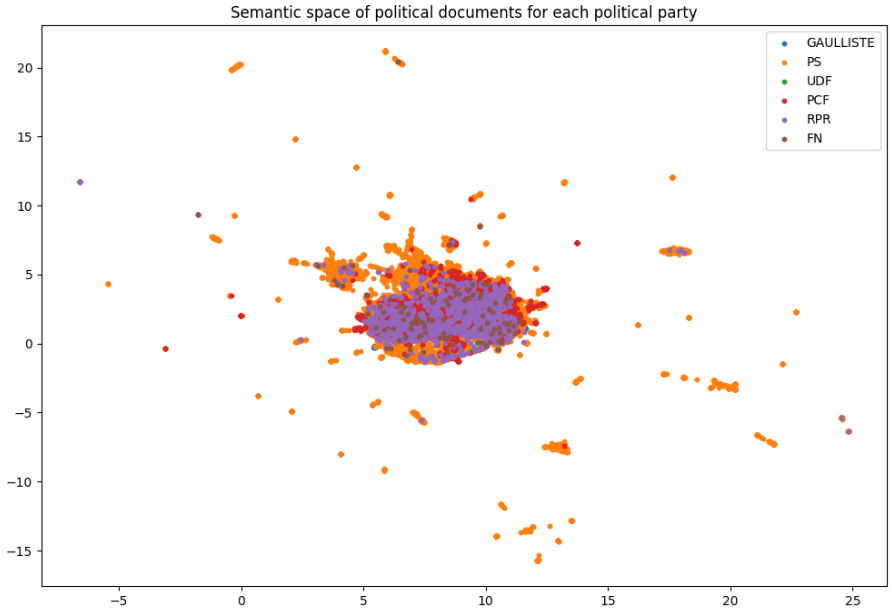

Figure 6: 2D semantic embedding colored by political party

5.3 Results and Analysis

The results show that:

- Documents cluster strongly in the center (for gender and for political party), but it seems independent of the class
- Thus, some overlap exists between centrist and allied parties
- Outliers correspond to OCR noise or atypical political messaging and can be seen easily, although there are not much of them

These findings sadly cannot confirm that embedding-based representations capture meaningful political structure in historical documents.

Conclusion

This quite little study of ours demonstrates that modern NLP techniques can effectively structure and analyze large corpuses of political texts without much supervision. By combining sentence embeddings, dimensionality reduction, and clustering, we tried to find latent ideological structures and identify anomalous documents of a large corpus.

However, the weakness of our classifiers do not allow us to lead relevant research on the matter, and the fact that the majority of our dataset belonged to Parti Socialiste has surely diminished the interest of trying to visualize the political party in a 2D semantic embedding space. Also, a larger dataset would have been preferable.

For future work, one might be thinking of including supervised classification of political ideology and temporal analysis of political speech evolution, as we remark that one thing missing form our study is the use of dates or any kind of temporal dimension into the visualizations.

References

[1] Semantic Similarity in NLP, Wikipedia.

[2] Reimers & Gurevych, Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.

[3] McInnes et al., UMAP: Uniform Manifold Approximation and Projection, 2018.